



Lecture # 20-21 Relational Database Design Algorithms

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Chapter Outline

- Designing a Set of Relations
- Properties of Relational Decompositions
- Algorithms for Relational Database Schema
- Multivalued Dependencies and Fourth Normal Form



DESIGNING A SET OF RELATIONS (1)

The Approach of Relational Synthesis (Bottom-up Design) :

- Assumes that all possible functional dependencies are known.
- First constructs a minimal set of FDs
- Then applies algorithms that construct a target set of 3NF or BCNF relations.
- Additional criteria may be needed to ensure the the *set of relations* in a relational database are satisfactory (dependency preservation and non-additive join).



DESIGNING A SET OF RELATIONS (2)

Goals:

- Lossless join property (a must) – algorithm 1 tests for general losslessness
- Dependency preservation property – algorithm 3 decomposes a relation into BCNF components by sacrificing the dependency preservation.
- Additional normal forms
 - 4NF (based on multi-valued dependencies)
 - 5NF (based on join dependencies)



1. Properties of Relational Decompositions (1)

Relation Decomposition and Insufficiency of Normal Forms:

- **Universal Relation Schema:** a relation schema $R = \{A_1, A_2, \dots, A_n\}$ that includes all the attributes of the database.
- **Universal relation assumption:** every attribute name is unique.
- **Decomposition:** The process of decomposing the universal relation schema R into a set of relation schemas $D = \{R_1, R_2, \dots, R_m\}$ that will become the relational database schema by using the functional dependencies.



Properties of Relational Decompositions (2)

Relation Decomposition and Insufficiency of Normal Forms (cont.):

- **Attribute preservation condition:** Each attribute in R will appear in at least one relation schema R_i in the decomposition so that no attributes are “lost”.
- Another goal of decomposition is to have each individual relation R_i in the decomposition D be in BCNF or 3NF.
- Additional properties of decomposition are needed to prevent from generating spurious tuples



Properties of Relational Decompositions (3)

Dependency Preservation Property of a Decomposition :

Definition:

- Given a set of dependencies F on R , the **projection of F** on R_i , denoted by $p_{R_i}(F)$ where R_i is a subset of R , is the set of dependencies $X \rightarrow Y$ in F^+ such that the attributes in $X \cup Y$ are all contained in R_i . *Hence, the projection of F on each relation schema R_i in the decomposition D is the set of functional dependencies in F^+ , the closure of F , such that all their left- and right-hand-side attributes are in R_i .*



Properties of Relational Decompositions (4)

Dependency Preservation Property of a Decomposition (cont.):

- **Dependency Preservation Property**: a decomposition $D = \{R_1, R_2, \dots, R_m\}$ of R is **dependency-preserving** with respect to F if the union of the projections of F on each R_i in D is equivalent to F ; that is, $((\pi_{R_1}(F)) \cup \dots \cup (\pi_{R_m}(F)))^+ = F^+$

Claim 1: It is always possible to find a dependency-preserving decomposition D with respect to F such that each relation R_i in D is in 3NF.



Properties of Relational Decompositions (5)

Lossless (Non-additive) Join Property of a Decomposition:

Definition:

- **Lossless join property**: a decomposition $D = \{R_1, R_2, \dots, R_m\}$ of R has the **lossless (nonadditive) join property** with respect to the set of dependencies F on R if, for *every* relation state r of R that satisfies F , the following holds, where $*$ is the natural join of all the relations in D :

$$* (\pi_{R_1}(r), \dots, \pi_{R_m}(r)) = r$$

Note: The word loss in *lossless* refers to *loss of information*, not to loss of tuples. In fact, for “loss of information” a better term is “addition of spurious information”



Properties of Relational Decompositions (6)

Lossless (Non-additive) Join Property of a Decomposition (cont.):

Algorithm 1: Testing for Lossless Join Property

Input: A universal relation R , a decomposition $D = \{R_1, R_2, \dots, R_m\}$ of R , and a set F of functional dependencies.

1. Create an initial matrix S with one row i for each relation R_i in D , and one column j for each attribute A_j in R .
2. Set $S(i,j) := b_{ij}$ for all matrix entries. (* each b_{ij} is a distinct symbol associated with indices (i,j) *).
3. For each row i representing relation schema R_i
 {for each column j representing attribute A_j
 {if (relation R_i includes attribute A_j) then set $S(i,j) := a_j$ };};
(* each a_j is a distinct symbol associated with index (j) *)



Properties of Relational Decompositions (7)

Lossless (Non-additive) Join Property of a Decomposition (cont.):

Algorithm 1: Testing for Lossless Join Property (cont.)

4. Repeat the following loop until a *complete loop execution* results in no changes to S
 - {for each functional dependency $X \rightarrow Y$ in F
 - {for all rows in S which have the same symbols in the columns corresponding to attributes in X
 - {make the symbols in each column that correspond to an attribute in Y be the same in all these rows as follows: if any of the rows has an “ a ” symbol for the column, set the other rows to that *same* “ a ” symbol in the column. If no “ a ” symbol exists for the attribute in any of the rows, choose one of the “ b ” symbols that appear in one of the rows for the attribute and set the other rows to that same “ b ” symbol in the column ;};};
5. If a row is made up entirely of “ a ” symbols, then the decomposition has the lossless join property; otherwise it does not.

Properties of Relational Decompositions (8)

Lossless (nonadditive) join test for n -ary decompositions.

(a) Case 1: Decomposition of EMP_PROJ into EMP_PROJ1 and EMP_LOCS fails test. (b) A decomposition of EMP_PROJ that has the lossless join property.

- (a) $R = \{SSN, ENAME, PNUMBER, PNAME, PLOCATION, HOURS\}$ $D = \{R_1, R_2\}$
 $R_1 = \text{EMP_LOCS} = \{ENAME, PLOCATION\}$
 $R_2 = \text{EMP_PROJ1} = \{SSN, PNUMBER, HOURS, PNAME, PLOCATION\}$
 $F = \{SSN \rightarrow ENAME; PNUMBER \rightarrow \{PNAME, PLOCATION\}; \{SSN, PNUMBER\} \rightarrow HOURS\}$

	SSN	ENAME	PNUMBER	PNAME	PLOCATION	HOURS
R_1	b_{11}	a_2	b_{13}	b_{14}	a_5	b_{16}
R_2	a_1	b_{22}	a_3	a_4	a_5	a_6

(no changes to matrix after applying functional dependencies)

(b)

EMP		PROJECT			WORKS_ON		
SSN	ENAME	PNUMBER	PNAME	PLOCATION	SSN	PNUMBER	HOURS

Properties of Relational Decompositions (8)

Lossless (nonadditive)
join test for n -ary
decompositions.

(c) Case 2:
Decomposition of
EMP_PROJ into EMP,
PROJECT, and
WORKS_ON satisfies
test.

(c) $R = \{SSN, ENAME, PNUMBER, PNAME, PLOCATION, HOURS\}$ $D = \{R_1, R_2, R_3\}$
 $R_1 = EMP = \{SSN, ENAME\}$
 $R_2 = PROJ = \{PNUMBER, PNAME, PLOCATION\}$
 $R_3 = WORKS_ON = \{SSN, PNUMBER, HOURS\}$

$F = \{SSN \rightarrow \{ENAME, PNUMBER\}; PNUMBER \rightarrow \{PNAME, PLOCATION\}; \{SSN, PNUMBER\} \rightarrow HOURS\}$

	SSN	ENAME	PNUMBER	PNAME	PLOCATION	HOURS
R_1	a_1	a_2	b_{13}	b_{14}	b_{15}	b_{16}
R_2	b_{21}	b_{22}	a_3	a_4	a_5	b_{26}
R_3	a_1	b_{32}	a_3	b_{34}	b_{35}	a_6

(original matrix S at start of algorithm)

	SSN	ENAME	PNUMBER	PNAME	PLOCATION	HOURS
R_1	a_1	a_2	b_{13}	b_{14}	b_{15}	b_{16}
R_2	b_{21}	b_{22}	a_3	a_4	a_5	b_{26}
R_3	a_1	b_{32} a_2	a_3	b_{34} a_4	b_{35} a_5	a_6

(matrix S after applying the first two functional dependencies -
last row is all "a" symbols, so we stop)



Properties of Relational Decompositions (9)

Testing Binary Decompositions for Lossless Join Property:

- **Binary Decomposition:** decomposition of a relation R into two relations.
- **PROPERTY LJ1 (lossless join test for binary decompositions):** A decomposition $D = \{R_1, R_2\}$ of R has the lossless join property with respect to a set of functional dependencies F on R *if and only if* either
 - The f.d. $((R_1 \cap R_2) \rightarrow (R_1 - R_2))$ is in F^+ , or
 - The f.d. $((R_1 \cap R_2) \rightarrow (R_2 - R_1))$ is in F^+ .



Properties of Relational Decompositions (10)

Successive Lossless Join Decomposition:

- **Claim 2 (Preservation of non-additivity in successive decompositions):**

If a decomposition $D = \{R_1, R_2, \dots, R_m\}$ of R has the lossless (non-additive) join property with respect to a set of functional dependencies F on R , and if a decomposition $D_i = \{Q_1, Q_2, \dots, Q_k\}$ of R_i has the lossless (non-additive) join property with respect to the projection of F on R_i , then the decomposition $D_2 = \{R_1, R_2, \dots, R_{i-1}, Q_1, Q_2, \dots, Q_k, R_{i+1}, \dots, R_m\}$ of R has the non-additive join property with respect to F .



2. Algorithms for Relational Database Schema Design (1)

Algorithm 2: Relational Synthesis into 3NF with Dependency Preservation (*Relational Synthesis Algorithm*)

Input: A universal relation R and a set of functional dependencies F on the attributes of R .

1. Find a minimal cover G for F ;
2. For each left-hand-side X of a functional dependency that appears in G , create a relation schema in D with attributes $\{X \cup \{A_1\} \cup \{A_2\} \dots \cup \{A_k\}\}$, where $X \rightarrow A_1, X \rightarrow A_2, \dots, X \rightarrow A_k$ are the only dependencies in G with X as left-hand-side (X is the *key* of this relation) ;
3. Place any remaining attributes (that have not been placed in any relation) in a single relation schema to ensure the attribute preservation property.

Claim 3: *Every relation schema created by Algorithm 2 is in 3NF.*



Algorithms for Relational Database Schema Design (2)

Algorithm 3: Relational Decomposition into BCNF with Lossless (non-additive) join property

Input: A universal relation R and a set of functional dependencies F on the attributes of R .

1. Set $D := \{R\}$;
2. While there is a relation schema Q in D that is not in BCNF do {
 choose a relation schema Q in D that is not in BCNF;
 find a functional dependency $X \rightarrow Y$ in Q that violates BCNF;
 replace Q in D by two relation schemas $(Q - Y)$ and $(X \cup Y)$;
};

Assumption: *No null values are allowed for the join attributes.*



Algorithms for Relational Database Schema Design (3)

Algorithm 4 Relational Synthesis into 3NF with Dependency Preservation and Lossless (Non-Additive) Join Property

Input: A universal relation R and a set of functional dependencies F on the attributes of R .

1. Find a minimal cover G for F (*Use Algorithm 10.2*).
2. For each left-hand-side X of a functional dependency that appears in G , create a relation schema in D with attributes $\{X \cup \{A_1\} \cup \{A_2\} \dots \cup \{A_k\}\}$, where $X \rightarrow A_1, X \rightarrow A_2, \dots, X \rightarrow A_k$ are the only dependencies in G with X as left-hand-side (X is the *key* of this relation).
3. If none of the relation schemas in D contains a key of R , then create one more relation schema in D that contains attributes that form a key of R . (*Use Algorithm 4a to find the key of R*)



Algorithms for Relational Database Schema Design (4)

Algorithm 4a Finding a Key K for R Given a set F of Functional Dependencies

Input: A universal relation R and a set of functional dependencies F on the attributes of R .

1. Set $K := R$.
2. For each attribute A in K {
 compute $(K - A)^+$ with respect to F ;
 If $(K - A)^+$ contains all the attributes in R ,
 then set $K := K - \{A\}$; }



Algorithms for Relational Database Schema Design (7)

Discussion of Normalization Algorithms:

Problems:

- The database designer must first specify *all* the relevant functional dependencies among the database attributes.
- These algorithms are *not deterministic* in general.
- It is not always possible to find a decomposition into relation schemas that preserves dependencies and allows each relation schema in the decomposition to be in BCNF (instead of 3NF as in Algorithm 4).

Algorithms for Relational Database Schema Design

(8)

Table Summary of some of the algorithms discussed above

Algorit hm	Input	Output	Properties/Purp ose	Remarks
1	A decomposition D of R and a set F of functional dependencies	Boolean result: yes or no for lossless join property	Testing for non-additive join decomposition	See the simpler test for binary decompositions
2	Set of functional dependencies F	A set of relations in 3NF	Dependency preservation	No guarantee of satisfying lossless join property
3	Set of functional dependencies F	A set of relations in BCNF	Lossless join decomposition	No guarantee of dependency preservation
.4	Set of functional dependencies F	A set of relations in 3NF	Lossless join and dependency preserving decomposition	May not achieve BCNF
4a	Relation schema R with a set of functional dependencies F	Key K of R	To find a key K (which is a subset of R)	The entire relation R is always a default superkey



Multivalued Dependencies and Fourth Normal Form (1)

Inference Rules for Functional and Multivalued Dependencies:

IR1 (reflexive rule for FDs): If $X \supseteq Y$, then $X \rightarrow Y$.

IR2 (augmentation rule for FDs): $\{X \rightarrow Y\} \mid = XZ \rightarrow YZ$.

IR3 (transitive rule for FDs): $\{X \rightarrow Y, Y \rightarrow Z\} \mid = X \rightarrow Z$.

IR4 (complementation rule for MVDs): $\{X \twoheadrightarrow Y\} \mid = X \twoheadrightarrow (R - (X \cup Y))$.

IR5 (augmentation rule for MVDs): If $X \twoheadrightarrow Y$ and $W \supseteq Z$ then $WX \twoheadrightarrow YZ$.

IR6 (transitive rule for MVDs): $\{X \twoheadrightarrow Y, Y \twoheadrightarrow Z\} \mid = X \twoheadrightarrow (Z \sqcup Y)$.

IR7 (replication rule for FD to MVD): $\{X \rightarrow Y\} \mid = X \twoheadrightarrow Y$.

IR8 (coalescence rule for FDs and MVDs): If $X \twoheadrightarrow Y$ and there exists W with the properties that (a) $W \cap Y$ is empty, (b) $W \rightarrow Z$, and (c) $Y \supseteq Z$, then $X \rightarrow Z$.



Multivalued Dependencies and Fourth Normal Form (2)

Lossless (Non-additive) Join Decomposition into 4NF Relations:

- **PROPERTY LJ1'**

The relation schemas R_1 and R_2 form a lossless (non-additive) join decomposition of R with respect to a set F of functional *and* multivalued dependencies if and only if

$$(R_1 \cap R_2) \twoheadrightarrow (R_1 - R_2)$$

or by symmetry, if and only if

$$(R_1 \cap R_2) \twoheadrightarrow (R_2 - R_1).$$



Multivalued Dependencies and Fourth Normal Form (3)

Algorithm 5: Relational decomposition into 4NF relations with non-additive join property

Input: A universal relation R and a set of functional and multivalued dependencies F .

1. Set $D := \{ R \}$;
2. While there is a relation schema Q in D that is not in 4NF do
 - { choose a relation schema Q in D that is not in 4NF;
 - find a nontrivial MVD $X \twoheadrightarrow Y$ in Q that violates 4NF;
 - replace Q in D by two relation schemas $(Q - Y)$ and $(X \cup Y)$;
 - };



Summary

- Discussed briefly the bottom up design approach
- Properties of Relational Decompositions discussed
 - Attribute preservation
 - Dependency preservation
 - Lossless join
- Algorithms for Relational Database Schema
 - Algorithm 1: Testing for Lossless Join Property
 - Algorithm 2: Relational Synthesis into 3NF with Dependency Preservation (*Relational Synthesis Algorithm*)
 - Algorithm 3: Relational Decomposition into BCNF with Lossless (non-additive) join property
 - Algorithm 4 Relational Synthesis into 3NF with Dependency Preservation and Lossless (Non-Additive) Join Property
 - Algorithm 4a Finding a Key K for R Given a set F of Functional Dependencies
 - Algorithm 5: Relational decomposition into 4NF relations with non-additive join property
- Multivalued Dependencies and Fourth Normal -properties